Department of Physics, FNSPE, CTU in Prague

02YELMA - Electricity and Magnetism

Final Exam

22.6.2026 / 14:00 - 16:30

Q1: A resistor (R), an inductor (L), a capacitor (C) and a switch are connected in series. The capacitor is initially charged to a voltage and the switch is closed at time $t = 0$.

(a) Write down the second-order differential equation governing the current $I(t)$ in the circuit (eliminate Q). — *0.3 points*

(b) In the underdamped case, how long does it take for the amplitude (envelope) of the oscillating current to fall to half its maximum value (*0.3 points*)? State the condition on R, L, and C for the underdamped regime (*0.1 points*).

(c) Now suppose the circuit is driven sinusoidally at resonance. Given that the quality factor of the circuit is defined as $Q = 2\pi \frac{\text{maximum energy stored}}{\text{energy dissipated per cycle}}$, find the quality factor. — *0.3 points*

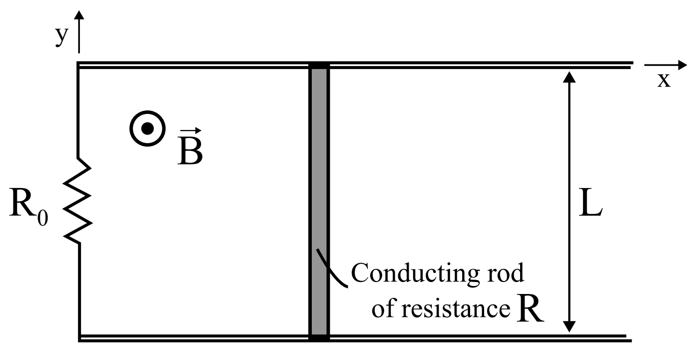
Q2: A long coaxial cable consists of two concentric conducting cylindrical shells with radii a and b ($a < b$) separated by vacuum. It is driven by a time-dependent current $I(t) = I_0 \cos(\omega t)$ that flows along the inner conductor in the positive z-direction and returns along the outer conductor in the negative z-direction. Assume cylindrical symmetry and neglect end effects (quasi-magnetostatic regime).

(a) Determine the magnetic field $\vec{B}(r, t)$ in the region $a < r < b$ (*0.4 points*) and $r > b$ (*0.2 points*).

(b) Calculate the instantaneous magnetic energy stored per unit length and inductance per unit length of the cable. — *0.4 points*

Q3: A long solenoid is at rest in frame S' and produces a uniform magnetic field inside its volume, $\vec{E}' = 0$, $\vec{B}' = B_0\,\hat{z}'$ where B_0 is a positive constant. Frame S' moves with constant velocity $\vec{v} = v\,\hat{x}$ relative to the laboratory frame S (i.e., perpendicular to the solenoid's axis). The axes of the coordinate systems are aligned. Using the field transformation laws, determine the electric field $\vec{E}$ (*0.5 point*) and magnetic field $\vec{B}$ (*0.5 points*) measured inside the solenoid in the laboratory frame S. *Given:* $E_\parallel = E'_\parallel$, $E_\perp = \gamma(E'_\perp - \vec{v} \times \vec{B}')$, $B_\parallel = B'_\parallel$, and $B_\perp = \gamma(B'_\perp + \frac{1}{c^2}\vec{v} \times \vec{E}')$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$ is the Lorentz factor and directions $\parallel$ and $\perp$ are defined with respect to the velocity $\vec{v}$.

Q4: A conducting rod of length L and resistance R can slide without friction on two horizontal conducting rails. The rails are connected through a resistor of resistance R_0, forming a closed circuit. The mass of the rod is m. Initially, the rod is at position $x = x_0$ and is at rest $v(0) = 0$. A spatially uniform magnetic field $\vec{B}(t) = B_0 e^{-\alpha t}\hat{z}$ is applied, where B_0 and α are positive constants.



(a) Determine the magnitude and direction of the induced current $I(t)$ in the circuit. — *0.4 points*

(b) Obtain the differential equation governing the motion of the rod in terms of its position and its derivatives. — *0.4 points*

(c) Calculate the instantaneous power dissipated in both resistors at time $t = 0$. — *0.2 points*